SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:		**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format:
     As a <user>, I want <feature>, so that <benefit>.
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Agile Sprint Planning & User Story Workshop

Project: AI-Based Fake Identity & Document Screening System

1. Project Vision and Stakeholders

Vision

To build an AI-powered screening platform that automatically detects fake or manipulated identity documents and synthetic identities in real time, enabling organizations to onboard users safely, comply with KYC/AML regulations, and reduce fraud losses.

Objectives

- Detect forged, altered, and AI-generated identity documents with high accuracy.
- Match the document holder's face to the ID photo with liveness verification.
- Provide explainable risk scores and audit trails for compliance teams.
- Integrate easily with existing onboarding and KYC workflows via APIs.
- Continuously improve detection models using new fraud patterns.

Key Stakeholders

- Product Owner / Business Sponsor – defines priorities, ROI, and compliance needs.
- Compliance & Risk Team – ensures KYC/AML regulatory alignment.
- End Users (Operations/Fraud Analysts) – review flagged cases and make decisions.
- Developers & Data Scientists – build and maintain AI models and platform.
- QA / Test Engineers – validate accuracy, performance, and security.
- Customers / Partner Organizations – consume the API or dashboard for onboarding.
- Legal & Privacy Officers – oversee data protection and consent handling.

2. Epics and User Stories

Epic 1: Document Capture & Pre-Processing

- US-1.1 As a user, I want to upload or capture an ID document via web/mobile, so that the system can start verification.
- US-1.2 As a system, I want to validate image quality (blur, glare, completeness), so that only usable images proceed to analysis.

- Given a counterfeit ID with mismatched fonts/security features
  When the authenticity engine analyzes it
  Then the system flags it as “suspected fake” with specific reason codes (e.g., font mismatch, missing hologram pattern).

US-2.2: AI-Generated/Altered Document Detection

- Given an AI-generated ID with inconsistent textures and compression artifacts
  When the synthetic-document detector runs
  Then the system labels it as “likely AI-generated” with a risk score above the configured threshold.
- Given a digitally altered ID where photo or text has been edited
  When the manipulation detector runs
  Then the system detects editing artifacts and raises an alert with evidence snapshots.

US-3.1: Face Match with ID Photo

- Given a clear selfie and a valid ID photo
  When the face-matching engine runs
  Then the system returns a match score and a pass/fail decision based on the configured threshold.
- Given two different people (selfie vs ID)
  When the face-matching engine runs
  Then the system returns a low match score and fails the biometric check.

US-3.2: Liveness Detection

- Given a live human in front of the camera
  When the liveness check runs
  Then the system passes the liveness test.
- Given a printed photo or replayed video presented to the camera
  When the liveness check runs
  Then the system detects spoofing and fails the liveness test.

US-4.1: Risk Score & Reason Codes

- Given all checks (document, biometric, liveness) are complete
  When the decision engine aggregates results
  Then the system produces an overall risk score and a list of reason codes explaining the decision.
- Given a high-risk case (e.g., fake document + failed liveness)
  When the decision engine runs
  Then the case is auto-flagged for manual review with “high risk” status.

US-4.2: Audit Logs

- Given a verification session is completed
  When the system finalizes the case
  Then it stores immutable logs including timestamps, inputs (hashed), model versions, scores, and decisions.
- Given an auditor requests logs for a specific user ID and date range
  When they query the audit module
  Then the system returns all relevant verification records in a downloadable format.

4. Prioritized Product Backlog

1. US-1.1 – Document upload/capture (foundational).
2. US-1.2 – Image quality validation (blocks downstream accuracy).
3. US-2.1 – Document authenticity check (core fraud detection).
4. US-3.1 – Face match with ID photo (core identity proofing).
5. US-3.2 – Liveness detection (anti-spoofing).
6. US-2.2 – AI-generated/alteration detection (advanced fraud).
7. US-4.1 – Risk score & reason codes (decision support).
8. US-5.1 – Secure APIs/SDKs (integration).
9. US-4.2 – Audit logs (compliance).
10. US-5.2 – RBAC & admin controls (security & governance).

5. Sprint 1 Backlog (First 2-Week Sprint)

Sprint 1 Focus: Enable basic document capture, quality checks, and initial authenticity + face match in a minimal end-to-end flow.

Selected User Stories for Sprint 1

- US-1.1 – Upload/capture ID document
- US-1.2 – Image quality validation
- US-2.1 – Document authenticity check (basic rules + template matching)
- US-3.1 – Face match with ID photo (basic model)
- US-4.1 – Risk score & reason codes (simple aggregation)

Breakdown into Tasks (examples)

US-1.1 Tasks

- Design upload UI (web + mobile mockups).
- Implement file upload API with size/type validation.
- Add client-side preview and error handling.
- Write unit and integration tests for upload flow.

US-1.2 Tasks

- Integrate image quality library (blur, glare, crop detection).
- Define quality thresholds per document type.
- Implement retry/recapture flow in UI.
- Test with varied image samples.

US-2.1 Tasks

- Integrate document template database for supported countries.
- Implement security feature checks (font, layout, barcode/MRZ consistency).
- Build basic anomaly scoring logic.
- Create test dataset of genuine vs obvious fakes.

US-3.1 Tasks

- Integrate face detection and embedding model.
- Implement face matching between selfie and ID photo.
- Define match threshold and tune on sample data.
- Add result visualization (match score) in UI.

US-4.1 Tasks

- Design risk score formula (weights for document, biometric).
- Implement decision engine (pass/fail + risk bands).
- Generate reason codes from each module.
- Display risk summary in results screen and API response.

6. Effort Estimation

Using Planning Poker with a Fibonacci scale (1, 2, 3, 5, 8, 13), assuming a mid-sized team familiar with ML APIs and web/mobile dev.

- US-1.1 – 5 points
  - Rationale: Standard upload feature but needs multi-platform support and robust validation.